

Title

Your name here

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Exercise 1 [2]

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Tools and resources used: On Exercise 1, I used ChatGPT to help me figure out how to [...]. On Choice Exercise 7, I watched a YouTube video to [...].

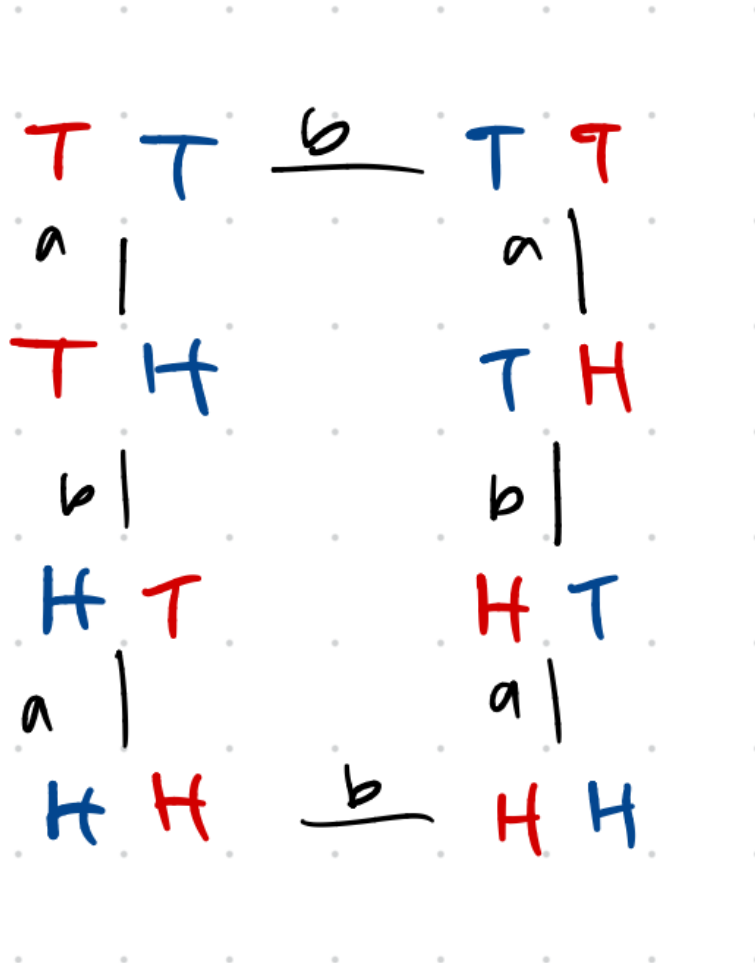
- Exercise 1 Points attempted: 2. Time spent: 5 minutes. Difficulty: 1/10. Self-assessed score: 2. Questions/Feedback: None.
- Exercise 2 Points attempted: 1. Time spent: 2 minutes. Difficulty: 1/10. Self-assessed score: 1. Questions/Feedback: None.
- Exercise 3 Points attempted: 6. Time spent: 40 minutes. Difficulty: 3/10. Self-assessed score: 6. Questions/Feedback: Did I do the problem correctly and as intended?
- Exercise 4 Points attempted: 5. Time spent: 35 minutes. Difficulty: 4/10. Self-assessed score: 5. Questions/Feedback: None.
- Exercise 5 Points attempted: 1. Time spent: 5 minutes. Difficulty: 1/10. Self-assessed score: 1. Questions/Feedback: None.
- Exercise 8 Points attempted: 7. Time spent: 40 minutes. Difficulty: 4/10. Self-assessed score: 7. Questions/Feedback: Mathematical correctness.
- Exercise 10 Points attempted: 3. Time spent: 10 minutes. Difficulty: 2/10. Self-assessed score: 3. Questions/Feedback: Mathematical correctness.

Exercise 2 [1]

Done! Posted an image of a marker with rotational symmetry.

Exercise 3 [6]

1. The coins have 8 symmetries. That is because there are 2^2 configurations without swapping places. When allowing swapping places, we get $2 \times 2^2 = 8$.
2. First, I chose the symmetry of flipping the *left*, denoted as f_L , coin over. Applying this symmetry twice gets us back to id. However, when we allow the swapping of places, denoted s , we can actually get to all states. We do not actually need an explicit 'flip right' symmetry, since it is actually equal to $f_L \circ s$.
3. In the diagram, let $a = f_L$ and $b = s$.



4. I used 2 generators.
5. This group is not commutative. This is because $f_L \circ s \neq s \circ f_L$. Swapping the coins then flipping the left is not the same as flipping the left then swapping.

Exercise 4 [5]

1. First, we write down the numbers in order to see 123456. Now, lets say we want to swap some things around, so we move some things around. Imagine that we move 1 to 2 and 2 to 1. We would get 213456. If we want to make some more complicated moves, we can actually send any number anywhere! So, $\pi = 152634$ would send 1 to 1 (unchanged), 2 to 5, 3 to 2, 4 to 6, 5 to 3, and 6 to 4.
2. We would get $\pi \circ \pi = 135426$.
3. We would get $(\pi \circ \pi) \circ \pi = 123654$.
4. Since $\pi^3 = 123654$, which is really just swapping 4 and 6, $\pi^3 \circ \pi^3 = \pi^6 = 123456$. So, π has an order of 6.
5. The right inverse of π is $\sigma = 135624$.
6. The left inverse of π is $\sigma = 135624$.
7. A word with order 1 is just any of the identity permutations. For example, 1234. For a word with order 2, we have 21. A general formula for constructing n order words is to just have n items, and for the word to be $\pi = 2345\dots n1$. This will “cycle” the items. So for order 4, we have 2341.

Exercise 5 [1]

1. The symmetry $r * h$ is a 120 degree rotation followed by a horizontal reflection.
2. The symmetry $r \star h$ is a 120 degree rotation after a horizontal reflection.
3. The function $f \circ g$ behaves like f after h .

Exercise 8 [7]

- [1] Write the word $2716543 \in S_7$ in cycle notation.

The word $2716543 \in S_7$ is equal to $(2731)(64)$ in cycle notation.

- [1] Write the permutation $(3)(514)(726)(8) \in S_8$ as a word.

The permutation $(514)(726) \in S_8$ is equal to the word 46351728.

- [1] Let $\pi = (514)(726)$ and let $\sigma = (618)(45)$. Write $\pi * \sigma$ in cycle notation, where $*$ means "followed by"?

Let $\pi = (514)(726)$ and let $\sigma = (618)(45)$.

$$\pi * \sigma = (586721).$$

- [1] Recall that the order of an element is the smallest positive number of times we have to apply it to get back to the identity. The order of the element $\tau = (123)(456)(7) \in S_7$ is 3 because $\tau \neq e \neq \tau^2$, but $\tau^3 = e$. What is the order of the element $(51)(632) \in S_7$?

The order of the element $\tau = (123)(456)(7) \in S_7$ is 3 because $\tau \neq e \neq \tau^2$, but $\tau^3 = e$. What is the order of the element $(51)(632) \in S_7$?

- [1] Can you find an element in S_5 that has order > 5 ? If so, give an example (written in cycle notation). If not, explain why not.

Yes. The element $(12)(345)$ has an order of 6.

- [2] What is the greatest order an element in S_7 can have? In S_{10} ?

The greatest order of an element in S_7 is (maybe) 12. This is done by making an element with two cycles, one with "size" 3 and order 3 and another cycle with "size" 4 and order 4. Then $\text{LCM}(3, 4) = 12$. I am actually unsure of whether this is the true max value for the order of elements in S_7 , since that would seemingly require checking many subcases.

For elements of S_{10} , I believe the max value is 30. In doing this problem, I decided to write a Python script to brute force all combos. The result ended up being 30.

```
from math import lcm
```

```
def get_distinct_sums(target, start=1):
    # Base case: if target is 0, we found a valid combination
    if target == 0:
        return [[]]

    results = []

    # Iterate from 'start' to the target
    for i in range(start, target + 1):
        # Recursively find combinations for the remaining value
        # We pass 'i + 1' to ensure the next number is strictly greater (distinct)
        sub_combinations = get_distinct_sums(target - i, i + 1)

        for combo in sub_combinations:
            results.append([i] + combo)

    return results

s7_sizes = get_distinct_sums(7)
```

```
s7_max_order = max(lcm(*size) for size in s7_sizes)
print(s7_max_order)
```

```
s10_sizes = get_distinct_sums(10)
s10_max_order = max(lcm(*size) for size in s10_sizes)
print(s10_max_order)
```

Choice Exercise 10 [3]

In class, we found that we can represent 120° rotation of the plane (one of the dihedral symmetries) with the matrix

$$R = \begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}.$$

1. [3] Write down the matrix that corresponds to f , a left-to-right flip, and then write down the matrices for the remaining four dihedral symmetries of the triangle D_3 .

We write the matrix F to represent the ‘left-to-right flip’ as:

$$F = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

First, we also define the id symmetry as the identity matrix:

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Then, we can construct all other symmetries of D_3 by composing F with R , meaning we take the matrix products of F and R . By convention, I will apply the F last.

So, we also have the symmetries R^2 , $F \times R$, and $F \times R^2$.